A synthetic DBA ring for KV tune-shift benchmarks and AI-assisted lattice design

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Abstract

A six-cell synthetic ring provides a common model for two tasks: comparison of space-charge tune shifts with the OPALX FFT2D5 solver, and evaluation of AI-assisted lattice reconstruction and optimization. A quadratic six-dimensional Hamiltonian specifies the external linear optics, including dispersion and longitudinal transport. A matched coasting Kapchinskij–Vladimirskij (KV) distribution supplies analytic transverse self-fields; numerical envelope matching and phase integration determine the incoherent tunes as functions of current and charge. This note defines the canonical conventions, cell parameters, benchmark operating points, and validation procedure. The machine-design task first reconstructs the ring from its cell and then adjusts the cell to prescribed tune targets. The reference calculations do not constitute an end-to-end validation of FFT2D5 tracking.

1 Purpose and model scope

The ring uses six identical cells, each containing two sector bends, two drifts, and one quadrupole. Its double-bend cell geometry motivates the DBA label, but the selected lattice is deliberately non-achromatic. The same geometry and focusing coefficients define both the space-charge benchmark and the machine-design task.

The external model assumes hard-edge fields, paraxial motion, first-order momentum deviations, and a fixed reference energy. It excludes fringe fields, chromatic nonlinearities, radio-frequency (RF) fields, radiation, and errors. The Hamiltonian is complete within this quadratic model; it does not describe exact nonlinear motion in magnetic fields. The KV comparison further assumes zero momentum spread, a longitudinally uniform coasting beam, and locally straight transverse space-charge forces.

Section 2 defines the external Hamiltonian and cell. Section 3 specifies the KV reference, and Section 4 describes the comparison with FFT2D5, the OPALX 2.5-dimensional space-charge solver. Section 5 defines the reconstruction and optimization tasks. File paths are relative to the project root, one directory above `TeX/`.

2 External Hamiltonian and ring construction

2.1 Canonical coordinates

The independent variable is the reference arc length s, measured in metres. A prime denotes differentiation with respect to s. The canonical state comprises three conjugate pairs,

$$\mathbf{u} = (x, p_x, y, p_y, \zeta, \eta)^{\mathsf{T}}. \tag{2.1}$$

The transverse coordinates x and y are measured in metres, and the normalized canonical momenta are

$$p_x = \frac{P_x}{p_0}, \qquad p_y = \frac{P_y}{p_0}, \tag{2.2}$$

where p_0 is the reference momentum. The ideal-magnet gauge used here has zero transverse vector potential. To linear order, p_x and p_y therefore equal the transverse slopes.

The longitudinal pair records arrival time and energy deviation:

$$\zeta = \beta_0 c \left(\frac{s}{\beta_0 c} - t \right), \qquad \eta = \frac{E - E_0}{\beta_0^2 E_0}. \tag{2.3}$$

Here t is the arrival time, E_0 is the reference total energy, and β_0 and γ_0 are the reference relativistic factors for 2.5 MeV protons. The coordinate ζ has units of metres and is positive for early arrival; η is dimensionless and equals the relative momentum deviation $\Delta p/p_0$ only to first order.

OPALX normalizes particle momentum by mc rather than p_0. For an on-momentum proton, the corresponding transverse conversion is

$$\frac{P_x}{mc} = \beta_0 \gamma_0 p_x, \tag{2.4}$$

with the analogous expression in the vertical plane.

The local horizontal coordinate points outward in a bend, and the local vertical coordinate points along y. At the ring entrance, positive curvature h bends the reference trajectory towards negative laboratory X. In the laboratory (X, Z) plane, the tangent and orientation satisfy

$$\hat{\mathbf{t}} = (\sin\theta, \cos\theta), \qquad \theta' = -h. \tag{2.5}$$

The entry and exit placements in `model/machine.json` follow this convention and match the copied baseline input deck.

2.2 Hamiltonian and exact linear maps

With these conventions, the scaled, dimensionless external Hamiltonian is

$$H_{\mathrm{ext}}(\mathbf{u}; s) = \frac{p_x^2 + p_y^2}{2} + \frac{k_x(s)x^2 + k_y(s)y^2}{2} - h(s)x\eta + \frac{\eta^2}{2\gamma_0^2}. \tag{2.6}$$

The curvature h has units of m^{-1}, and the focusing coefficients k_x and k_y have units of m^{-2}. Hamilton's equations give

$$\begin{aligned} x' &= p_x, & p_x' &= -k_x x + h\eta, \\ y' &= p_y, & p_y' &= -k_y y, \\ \zeta' &= -hx + \eta/\gamma_0^2, & \eta' &= 0. \end{aligned} \tag{2.7}$$

The coefficients are constant within each element. The coordinates and canonical momenta remain continuous at hard edges; the model applies no thin-edge kicks. Each element conserves its autonomous Hamiltonian, whereas the explicitly s-dependent Hamiltonian of the complete ring is not a globally conserved scalar.

Let G denote the Hessian of H_{ext} with respect to $\mathbf{u}$, and let

$$\mathsf{J} = \mathrm{diag}(\mathsf{J}_2, \mathsf{J}_2, \mathsf{J}_2), \qquad \mathsf{J}_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{2.8}$$

The linear equations and exact map for an element of length L are

$$\mathbf{u}' = \mathsf{J}\mathsf{G}\mathbf{u}, \qquad \mathsf{M}(L) = \exp(L\mathsf{J}\mathsf{G}). \tag{2.9}$$

These maps are symplectic and include dispersion and longitudinal coupling. In beam order, the cell and ring maps are

$$\mathsf{M}_{\mathrm{cell}} = \mathsf{M}_{B2}\mathsf{M}_{D2}\mathsf{M}_{Q}\mathsf{M}_{D1}\mathsf{M}_{B1}, \qquad \mathsf{M}_{\mathrm{ring}} = \mathsf{M}_{\mathrm{cell}}^{6}. \tag{2.10}$$

The canonical Hamiltonian context is described in Ref. [2]; the coordinate conventions and signs of the present quadratic model are specified explicitly in Eqs. (2.1)–(2.10) and checked against closed-form sector-bend maps.

2.3 Cell parameters and zero-current optics

Table 1 specifies the ordered cell. Repeating it six times produces 30 elements, including twelve 30° sector bends of radius 2 m. The total bend angle is 2π, and the circumference is 25.76637061436 m. Figure 1 shows the reference geometry and element sequence in laboratory coordinates.

Table 1: Hard-edge cell parameters. The coefficients k_x and k_y describe physical focusing in Eq. (2.6); they are not OPALX input attributes. Elements appear in beam order.

Element	L [m]	h [m^{-1}]	k_x [m^{-2}]	k_y [m^{-2}]
$B1$	$\pi/3$	0.5	0.25	0
$D1$	1	0	0	0
Q	0.2	0	-2.84	$+2.84$
$D2$	1	0	0	0
$B2$	$\pi/3$	0.5	0.25	0

The proton OPALX input uses $K_1 = -2.84\,\mathrm{m}^{-2}$ to implement the quadrupole coefficients in Table 1. The historical electron benchmark uses the opposite sign. Interpreting the OPALX attribute K_1 therefore requires the particle and charge convention.

The full positive zero-current tunes are

$$Q_x(0) = 0.770107569133, \qquad Q_y(0) = 1.689536397020. \tag{2.11}$$

The principal fractional one-turn eigenphases quoted in the original benchmark, 0.229892431 and 0.310463603, are compatible with these tunes but do not specify their full positive values. The generator selects a cell phase in $(0, 2\pi)$ from the sign of the corresponding transverse map entry M_{12} and multiplies the phase by six. An independent KV phase integral verifies this branch at the baseline. Local optimization remains near that branch; a design with more than one oscillation per cell requires independent phase unwrapping along s.

The periodic horizontal dispersion at the entrance is approximately 9.337460094 m, with zero slope. The longitudinal motion drifts; the machine provides no longitudinal confinement.

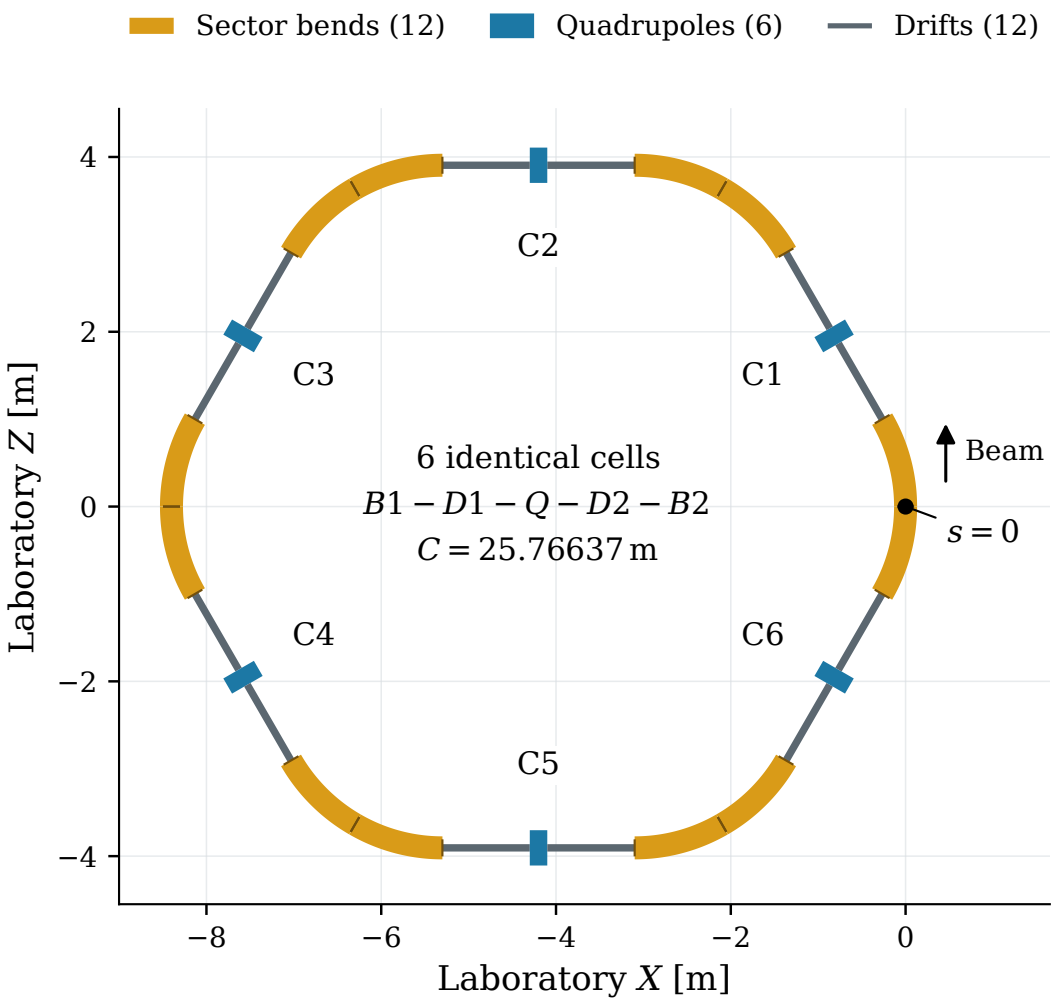


Figure 1: Reference geometry of the six-cell synthetic ring for the 2.5 MeV proton benchmark. Each cell follows the sequence $B1$–$D1$–Q–$D2$–$B2$; the complete ring contains twelve sector bends, twelve drifts, and six quadrupoles. The laboratory coordinates have equal scale. The arrow indicates the initial beam direction at $s = 0$, and short transverse strokes distinguish adjacent bends. Magnet widths are schematic and do not represent apertures.

3 Matched coasting KV reference

3.1 Envelope conventions and matching

The KV reference uses zero momentum spread, $\eta = 0$, and a longitudinally uniform coasting beam. An initially uniform coasting beam remains longitudinally uniform in this ideal model. The transverse edge radii a and b, measured in metres, relate to the root-mean-square (rms) beam sizes through

$$a = 2\sigma_x, \qquad b = 2\sigma_y. \tag{3.1}$$

The geometric edge emittances $\mathcal{E}_x$ and $\mathcal{E}_y$ (`Ex` and `Ey` in the original model notation) are four times the corresponding geometric rms emittances. The baseline rms emittance in each plane is $3 \times 10^{-6}\,\mathrm{m\,rad}$.

For positive proton current I, the Alfvén current and generalized perveance are

$$I_A = \frac{4\pi\epsilon_0 m_p c^3}{e}, \qquad K = \frac{2I}{I_A \beta_0^3 \gamma_0^3}, \tag{3.2}$$

where m_p is the proton mass, e is the elementary charge, and ϵ_0 is the vacuum permittivity. The coupled KV envelope equations are

$$\begin{aligned} a'' + k_x a - \frac{\mathcal{E}_x^2}{a^3} - \frac{2K}{a+b} &= 0, \\ b'' + k_y b - \frac{\mathcal{E}_y^2}{b^3} - \frac{2K}{a+b} &= 0. \end{aligned} \tag{3.3}$$

Reference [1] describes the KV envelope and perveance conventions. The calculation rematches the periodic envelopes at each current while holding the external focusing fixed. The self-field is analytic; envelope matching and phase integration are numerical.

3.2 Collective potential and particle tunes

Inside the uniform KV ellipse, the transverse collective contribution to the test-particle Hamiltonian is

$$H_{\mathrm{sc}} = -\frac{K}{a+b}\left(\frac{x^2}{a} + \frac{y^2}{b}\right). \tag{3.4}$$

The resulting effective focusing coefficients are

$$k_{x,\mathrm{eff}} = k_x - \frac{2K}{a(a+b)}, \qquad k_{y,\mathrm{eff}} = k_y - \frac{2K}{b(a+b)}. \tag{3.5}$$

Equation (3.4) describes an interior mean-field potential with self-consistently matched envelopes. Its quadratic form does not apply outside the ellipse. The model neither sums finite-particle pair interactions nor includes six-dimensional collective dynamics with longitudinal space charge. Self-consistency enters through Eq. (3.3); retaining the zero-current beam sizes as the current increases would change the model. Additive functions of s do not affect the transverse motion.

The KV distribution forms a uniform shell in normalized four-dimensional transverse phase space, with invariant

$$\left(\frac{x}{a}\right)^2 + \left(\frac{a p_x - a' x}{\mathcal{E}_x}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{b p_y - b' y}{\mathcal{E}_y}\right)^2 = 1. \tag{3.6}$$

A filled four-dimensional ball represents a different distribution. The full positive incoherent particle tunes follow from the matched envelopes:

$$Q_x(I) = \frac{6}{2\pi}\int_{\mathrm{cell}} \frac{\mathcal{E}_x}{a(s)^2}\,\mathrm{d}s, \qquad Q_y(I) = \frac{6}{2\pi}\int_{\mathrm{cell}} \frac{\mathcal{E}_y}{b(s)^2}\,\mathrm{d}s. \tag{3.7}$$

These tunes describe individual-particle motion rather than centroid motion or envelope modes. Finite amplitudes and curvature in an OPALX field tracker can produce departures from the paraxial, locally straight space-charge reference. Mesh and timestep convergence alone do not remove such model differences.

4 Comparison with FFT2D5

4.1 Charge normalization and field reference

A controlled comparison uses the beam parameters of Section 3, open transverse boundaries, and no RF fields or pipe-image forces. FFT2D5 uses laboratory-frame line charge and open transverse slice solves. Its longitudinal-field mode must be `FFT2D5LongitudinalFieldMode::None` for the purely transverse comparison; the input-deck spelling must be checked when constructing the run.

The current fixes the line charge λ, which in turn fixes the charge represented by a full ring or a finite segment:

$$\begin{aligned} \lambda &= \frac{I}{\beta_0 c}, & N_{\mathrm{ring}} &= \frac{\lambda C}{e}, \\ Q_{\mathrm{ring}} &= \lambda C, & Q_{\mathrm{segment}} &= \lambda L_{\mathrm{segment}}, \\ q_{\mathrm{macro}} &= \frac{Q_{\mathrm{segment}}}{N_{\mathrm{macro}}}. \end{aligned} \tag{4.1}$$

Here C is the circumference, L_{segment} is the represented segment length, and N_{macro} is the number of macroparticles in that segment. The line charge has units of coulombs per metre.

A full coasting ring requires particles distributed around the closed path with the appropriate local matched covariance at each s. A finite segment requires uniform interior slices and suitable longitudinal handling. The zero-length observation section in `kv-section.txt` cannot represent a charged coasting beam, and an isolated bunch with open boundaries describes a different physical model. Full-ring longitudinal wrapping and continuity across the path seam require verification before a ring comparison can be claimed.

Within the KV ellipse, the analytic laboratory electric field and the associated magnetic field are

$$\begin{aligned} E_x &= \frac{\lambda x}{\pi\epsilon_0 a(a+b)}, \quad E_y = \frac{\lambda y}{\pi\epsilon_0 b(a+b)}, \\ \mathbf{B} &= \frac{\beta_0}{c}\,\hat{\mathbf{t}} \times \mathbf{E}. \end{aligned} \tag{4.2}$$

The magnetic field reduces the net transverse force to

$$\mathbf{F}_\perp = \frac{e\mathbf{E}_\perp}{\gamma_0^2}. \tag{4.3}$$

The electric field in Eq. (4.2) already uses laboratory normalization and requires no additional factor of γ_0. Combining the field and force expressions gives

$$\frac{e(E_x/x)}{p_0\beta_0 c\gamma_0^2} = \frac{2K}{a(a+b)}, \tag{4.4}$$

which connects a local field-gradient check to the defocusing term in Eq. (3.5).

4.2 Reference operating points

Table 2 lists the reference currents and tunes. The tune shifts are defined by $\Delta Q_x = Q_x(I) - Q_x(0)$ and $\Delta Q_y = Q_y(I) - Q_y(0)$. The principal comparison point has $I = 10.399311\,\mathrm{mA}$ and $\Delta Q_y = -0.2$.

At the principal comparison point, the total ring charge is $12.26826245\,\mathrm{nC}$, corresponding to $7.65724714 \times 10^{10}$ protons. The entrance rms beam sizes are $\sigma_x = 6.042810\,\mathrm{mm}$ and $\sigma_y =$

Table 2: Coasting KV reference for 2.5 MeV protons with geometric rms emittances of 3×10^{-6} m rad in each plane. The full positive tunes Q_x and Q_y follow from the numerically matched periodic envelopes. The final row is envelope-unstable and serves only as a stress case.

I [mA]	Q_x	Q_y	ΔQ_x	ΔQ_y	Role
0	0.770108	1.689536	0	0	Bare optics
4.748770	0.623940	1.589536	−0.146167	−0.1	First collective comparison
10.399311	0.484005	1.489536	−0.286102	−0.2	Principal comparison
17.402677	0.359252	1.389536	−0.410855	−0.3	Envelope-unstable stress case

2.480302 mm. At the highest current in Table 2, the envelope multiplier is approximately 1.069761 per cell. That periodic solution is therefore unsuitable as a stable matched-beam reference.

Figure 2 expresses the analytic reference as a function of total coasting ring charge. The charge normalization is that of Eq. (4.1); it is not the charge of an arbitrary finite simulation segment. Crosses identify sampled periodic solutions with envelope instability. The scan brackets the onset of this instability between 15.740 nC and 16.424 nC; it does not locate an exact threshold.

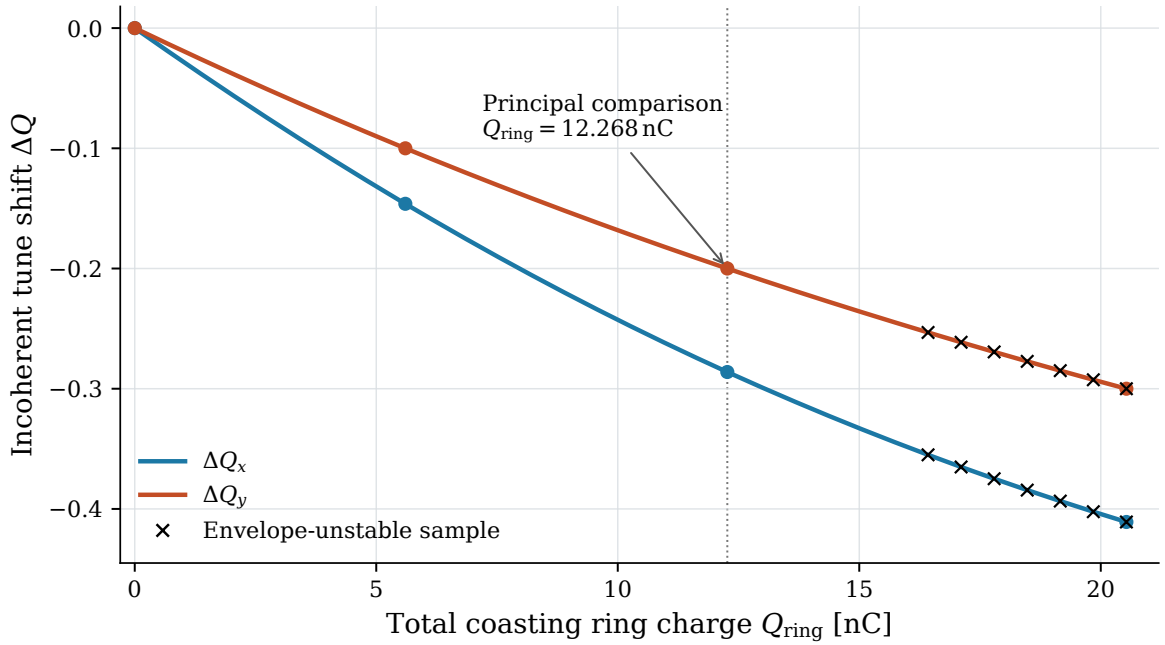


Figure 2: Incoherent KV tune shifts versus total coasting ring charge for the baseline 2.5 MeV proton lattice, with geometric rms emittances of 3×10^{-6} m rad in both planes and zero momentum spread. The curves join the saved periodic-envelope solutions for ΔQ_x and ΔQ_y; circles mark the operating points in Table 2, and crosses mark envelope-unstable samples. The dotted line identifies the principal comparison point, $Q_{\mathrm{ring}} = 12.268$ nC, where $\Delta Q_y = -0.2$. The self-fields are analytic, while envelope matching and phase integration are numerical. These curves are independent references, not FFT2D5 tracking results.

4.3 Solver validation procedure

The independent reference is available, but end-to-end OPALX FFT2D5 tracking remains to be validated. The comparison should proceed through the following stages:

(i) Establish convergence of the bare-ring particle and reference orbits and of the zero-charge tunes in OPALX. Historical tracking reports refer to earlier source revisions and do not establish the behavior of the current implementation.

(ii) Compare the FFT2D5 interior field gradients and net transverse force at a fixed matched section with Eqs. (4.2)–(4.4). Check deposited charge and line density as well as the fields, then repeat the comparison at a curved section to expose coordinate and reference-path errors.

(iii) Transport the correctly normalized matched source through one cell. Compare envelopes and projected moments with the reference at every element boundary.

(iv) Track the stable coasting operating points over many turns. Measure individual-particle tunes at a consistent observation plane, with the tune branches unfolded. Centroid motion or a principal fast-Fourier-transform peak alone does not provide this comparison. Include a zero-charge run with the same numerical settings.

(v) Quantify sensitivity to timestep, transverse mesh spacing and domain size, longitudinal slice spacing, macroparticle count, and spectral observation length. Hold the physical parameters fixed and report absolute tune errors and errors in ΔQ. Set acceptance tolerances from the convergence evidence.

Parameter studies use one Message Passing Interface (MPI) rank. Dedicated single-rank and multi-rank consistency checks belong to solver validation, rather than to the serial reference calculation. No OPALX or MPI solver validation has yet been completed for this standalone package.

5 AI-assisted reconstruction and optimization

5.1 Reconstruction task

The reconstruction task supplies an AI system with `model/cell.json` and the Hamiltonian specification in Section 2. The evaluator retains `model/machine.json` as the reference output when an independent reconstruction is required.

The system must derive the six Hamilton equations, construct the exact map of each element, compose the ordered cell map, and repeat the cell six times. It must also integrate the reference geometry and report all 30 placements, the cell and ring maps, circumference, total bend angle, periodic dispersion, full tunes, and symplectic residuals. The proton sign convention applies throughout. At zero current, the reconstructed tunes must reproduce the independent envelope phase integrals; principal one-turn eigenphases alone do not determine the full tune branches.

The baseline evaluator is `model/machine.py`, and the checks are implemented in `tests/test_machine.py`. The reconstruction targets are the circumference and cell geometry of Section 2, the full tunes in Eq. (2.11), and the entrance dispersion of 9.337460094 m with zero slope. The tests use near-roundoff tolerances for exact maps and geometry, and 2×10^{-10} for the comparison with numerically integrated envelope phases. These are model-verification tolerances, not acceptance criteria for particle-in-cell (PIC) tracking. The reconstruction must state the scope of the quadratic model separately from that of exact nonlinear field tracking.

5.2 Local tune optimization

After reconstruction, the design task adjusts the cell to illustrative full tune targets $(Q_x, Q_y) = (0.8, 1.7)$. These values define a demonstration rather than a proposed operating point or an optimal physics choice. The two free parameters are the physical quadrupole coefficient and the common length of the two drifts. The particle, reference energy, bend radii and angles, quadrupole length, and cell count remain fixed.

A candidate must have positive lengths, transversely stable cell optics, symplectic maps, and closed reference geometry. The circumference may change. Fixing it would generally prevent

these two parameters from matching two arbitrary tune targets; the design would then require an additional justified degree of freedom or a revised objective.

For each candidate, the system must export the cell specification, the complete machine, and a KV configuration with exactly the same element coefficients. It must then rematch the KV beam and recompute the charge-dependent tune shifts. Envelope-unstable solutions cannot serve as stable comparison points. Matching the bare tunes alone does not establish collective stability or robustness.

The script `model/optimize.py` provides a deterministic local reference optimizer for evaluating proposed designs. It does not demonstrate independent AI design and does not search globally across tune branches. It reports failure when local matching encounters an unstable or nonphysical cell, or when the calculation fails to converge.

The worked example in `designs/demo-qx0.8-qy1.7/` contains a complete machine with full tunes $(0.8, 1.7)$ and a compatible `kv-config.json` for the analytic scan. Evaluation should consider the achieved optics, geometry, collective stability, reproducibility, and stated design tradeoffs.

6 Reference implementation and validation status

The script `tune-shift-ring/analysis/run_model.py` regenerates the current scan, charge normalization, matched periodic envelopes, particle tunes, envelope-stability diagnostics, and local electric-field gradients. Each run writes its comparison targets to `fft2d5_reference.json` in a new result directory. These outputs provide the independent reference for the staged solver comparison of Section 4; they do not establish FFT2D5 tracking accuracy.

The machine representation and the matched KV calculation serve complementary purposes. The former specifies the external linear machine for reconstruction and design; the latter supplies the collective reference against which a numerical space-charge solver can be tested. Both retain the model assumptions and limits stated in Section 1.

The script `TeX/make_figures.py` regenerates the lattice and tune-shift figures from `model/machine.json` and the baseline scan selected by `REFERENCE_RUNS.json`. It verifies the charge normalization and geometry and records input hashes in `TeX/figures/provenance.json`. The vector PDF figures and PNG previews are stored in `TeX/figures/`.

References

[1] C. Prior, *Modelling with Space-Charge*, CERN Accelerator School lecture material. Available at `https://indico.cern.ch/event/221441/contributions/1524769/attachments/360572/502106/SpaceCharge_Modelling.pdf`.

[2] *Particle tracking in circular accelerators using the exact Hamiltonian.* Available at `https://s3.cern.ch/inspire-prod-files-7/74c88492a8d6a790563d145198c4813d`.